AAPG2022	DELICACY		PRC
Coordinated by:	Yannick CHASSAGNEUX	48 months	595 931 €
scientific evaluation panel: Axe E.6 : Technologies quantiques		CES: 47	

Graphene Quantum Dot in Dielectric Nano-Cavity

Summary table of persons involved in the project:

Partner	Name	First name	Current position	Role & responsibilities in the project	Involvement (person.month)
LPENS	CHASSAGNEUX	Yannick	CR	Coordinator, Fabrication of cavities, GQDs coupled to cavity Tasks 1.1 1.2 1.3 1.5 3.1 3.2 3.3 3.5	24 p.month
LPENS	VOISIN	Christophe	PR	GQDs coupled to cavity Tasks 1.4 3.2 3.4	12 p.month
LPENS	RAPISARDA	Federico	PhD	Cavity fabrication and characterization Tasks 1.1 1.2 1.3 1.4 1.5	12 p.month
LPENS	To be recruited		PhD	Coupling of GQD to cavity Tasks 3.1 3.2 3.3 3.4 3.5	36 p.month
LUMIN	LAURET	Jean-Sébastien	PR	Partner's Scientific leader Tasks 2.2 2.3 2.4	12 p.month
LUMIN	RONDIN	Loïc	MC	Single object spectroscopy Tasks 2.3 2.4	9 p.month
LUMIN	CASSETTE	Elsa	CR	Ensemble spectroscopy Tasks 2.2 2.3	6 p.month
LUMIN	To be recruited		PostDoc	Optical spectroscopy Tasks 2.2 2.3 2.4	12 p.month
NIMBE	CAMPIDELLI	Stephane	DR	Partner's Scientific leader Tasks 2.1 2.2	12 p.month
NIMBE	DERYCKE	Vincent	DR	Sample preparation, lithography Tasks 2.2 2.3	5 p.month
NIMBE	BANGA-KPAKO	Cynthia	PhD starting in oct. 2022	Chemistry: synthesis of GQDs Tasks 2.1 2.2	12 p.month
NIMBE	To be recruited		½ PhD	Chemistry: synthesis of GQDs Tasks 2.1 2.2	36 p.month

Any changes that have been made in the full proposal compared to the pre-proposal
No major change has been made between the full proposal compared to the pre-proposal.

I. Proposal's context, positioning and objective(s)

a. Objectives and research hypothesis

Optical cavities have been studied and used in many different contexts. Miniaturized cavities can be found in daily life laser diodes and kilometers long ones in gravitational wave experiments (LIGO, VIRGO). Optical cavities have undergone a great evolution in the sixties as an important element in the development of laser performance. They also emerged as fundamental tools to control light-matter interaction at an unprecedented level in the framework of cavity Quantum Electro Dynamics (cQED). Despite this long history, the design of new cavities is still an active research field, mainly in the context of linear or nonlinear light-matter interaction enhancement. At the level of a single (or few) emitter, light-matter interaction can be greatly enhanced with high-quality factor (Q) and small mode volume (V_{eff}) optical cavities, with several applications such as nano-laser [Lon02, Alt06], low-power nonlinear optics [Sol04, Noz10], sensitive detection of biological molecules [Arm07], cavity quantum electrodynamics [Eng07, Tie14]... The strength of these effects typically scales as the ratio Q_{eff}/V_{eff}, where the effective quality factor (Q_{eff}) is roughly given by the smallest of that of the cavity and that of the emitter. Q is typically imposed by the cavity losses, whereas V_{eff} is limited by the laws of diffraction. Hence, two main approaches have emerged to maximize cavity effects: 1- the plasmonic one where V_{eff} can be extremely small due to wavelength compression, but at the expense of Q because of the losses in metallic structures; 2 - the dielectric one where V_{eff} is limited to a fraction of λ^3 due to diffraction but with very high values of Q. The realization of an optical cavity combining both a strongly sub-wavelength mode volume and a high-quality factor is currently a missing achievement of photonics which would unlock the access to new regimes of light-matter interaction and open to new physics.

Therefore, the first objective of DELICACY is the demonstration of a nano-cavity combining high quality factor and deep sub-diffraction mode volume. To reach sub-diffraction mode volume a dielectric nano-fabricated structure that acts as a field "concentrator" will be integrated in the center of the cavity (cf Fig1).

In the framework of cQED, nano-cavities exhibit their full potential when coupled to a single quantum nano-emitter. The deterministic coupling of a nano-emitter to a cavity is usually a difficult task, which is sometimes solved on a trial and error basis, requiring the fabrication of a large number of devices. Here, we tackle this challenge on a deterministic approach: the cavity geometry is designed from the beginning to be as flexible as possible for efficient coupling to the emitter. To this end, the architecture of the cavity will be based on a flexible Fibered Fabry Perot (FFP) cavity. **The second objective of the DELICACY project is to efficiently couple a single nano-emitter to an innovative flexible cavity, combining high-quality factor, deeply sub-diffraction mode volume (*cf* Fig 1).**

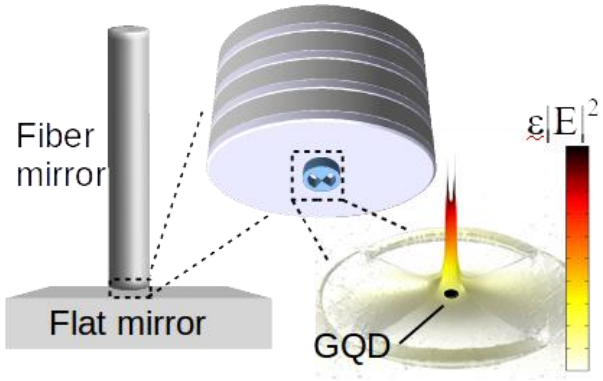


Fig 1: GQD coupled to a high Q - ultra-low mode volume cavity. The blue bow-tie structure is the dielectric "concentrator". One of the two mirrors is fabricated on the apex of an optical fiber to obtain the spectral and spatial matching.

Among the vast choice of possible quantum emitters, from cold atoms or ions to solid-state defects, graphene quantum dots (GQDs), synthesized via the bottom-up approach, appear to be a new player which combines a high brightness, room temperature operation together with great potential in terms of control of the intrinsic properties and adaptability towards targeted application, through the flexibility of the chemical synthesis.

To exemplify the power of this innovative system, we propose: 1- to demonstrate the realization of an efficient, room temperature tunable single photon source; 2- to perform highly sensitive absorption spectroscopy; 3- to reach the strong coupling regime.

b. Position of the project as it relates to the state of the art

Historically, two main routes have been followed for the coupling of a single nano-emitter to an optical cavity: the plasmonic one, where the electric field is concentrated using a metallic antenna leading to deep sub-wavelength mode volume $V_{eff} \sim 10^{-4}\ \lambda^3$, at the price of a relatively poor cavity quality factor: Q~10 (due to metallic losses) [Lal18]. The second route is based on purely dielectric cavities, with a Q factor up to Q~10^6, but with a mode volume limited by diffraction to $V_{eff} \sim \lambda^3$ [Lal18].

Until very recently, it was assumed that the mode volume for dielectric cavities was limited by diffraction: $V_{eff} \gtrsim (\lambda/2n)^3$ [Coc98]. In 2017, a theoretical paper [Cho17] questioned this picture, showing that the diffraction limit can be deeply overcome in dielectric cavities with volumes as small as $V_{eff} \sim 10^{-5}\ \lambda^3$, together with Q~$10^6$, by using an innovative electric field "concentrator" made of a bow-tie shaped sub-wavelength dielectric inclusion. This concentrator exploits the boundary conditions at dielectric interfaces in order to generate an electric field anomaly. The "concentrator" principle relies on the vectorial nature of the electro-magnetic fields and on the field's discontinuities on dielectric interfaces. A plane divided in angular dielectric sectors exhibits an electric field divergence in the center [Fis07]. In the case of a four quadrants structure, as planned to be used in DELICACY, the electric field diverges as $|E| \propto r^{(c_0-1)}$ where $c_0 = 2/\pi \arccos[(\varepsilon-1)/(\varepsilon+1)]$, ε is the permittivity of the dielectric medium and *r* is the radial coordinate (for an e-beam resist concentrator $|E| \propto 1/r^{0.25}$ and for a higher refractive index material such as TiO_2, the divergency goes as $|E| \propto 1/r^{0.5}$). Experimentally the E-field will remain finite: the smallest reachable r essentially reflects the radius of curvature of the tip of the dielectric bow-tie geometry. This implies that structures with either a nano-gap or a nano-bridge are equivalent in this respect; the fabrication challenge is thus to make the central region as small as possible, whatever a gap or a bridge. The outer shape of the concentrator is a ring of dielectric which is used to confine the mode laterally similarly to a micro-disk resonator. For a concentrator with bridge or gap width of 20 nm, we performed preliminary finite element simulations showing that a mode volume $V \sim 5.10^{-3}\ \lambda^3$ is reachable (cf Fig.2).

Partner LPENS has fabricated with e-beam lithography a "concentrator" on a quartz substrate to verify experimentally the feasibility of such devices (cf Fig 2). It turns out that the e-beam resist (HSQ:

Hydrogen silsesquioxane) leads to the smallest structure with an e-beam dose of 2640 µC/cm^2. An HSQ nano-bridge of 15 nm width has been obtained. AFM measurements show a rugosity of 0.8 nm, already below 1.8 nm required for a cavity with quality factor Q=10 000.

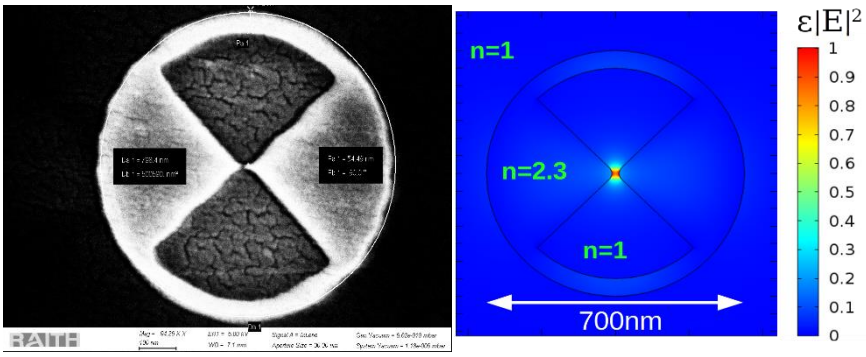


Fig 2. Left. SEM image of a fabricated "concentrator". Note that the apparent rugosity is due to a metallic coating that has been used for imaging purposes (to avoid charging effect). The white part corresponds to the e-beam resist. The central dielectric bridge is approximately 15 nm wide . Right. Preliminary simulation of the field distribution inside the "concentrator".

If properly used, this concentrator can transform a cavity with a ~λ^3 mode volume into a deep sub-wavelength mode volume cavity, while maintaining a high-quality factor. The resulting light matter interaction strength could be increased by several orders of magnitude. The seminal paper [Cho17] triggered many theoretical proposals (~ 100) to include similar electric field concentrators in various cavity geometries. To the best of our knowledge, only one experimental demonstration [Hu18] has been reported so far (V_{eff} ~ 1.6 10^{-4} λ^3, Q ~ 10^5), **still with no emitter inside the cavity**.

Experimentally, both plasmonic and dielectric approaches face challenging issues to couple efficiently a nano-emitter to the cavity. In fact, the spatial matching (the emitter should be placed in the maximum electric field), and the spectral matching (the cavity resonant frequency should match the emitter transition) have to be met simultaneously. A possible approach is to rely on random matching and search for the cavity-emitter pairs that match both conditions. In the case of high Q and small V cavities, the matching yield would be extremely limited (with V~λ^3 cavity, the success is already lower than 10^{-4} [Mac21]). Another approach is based on the prior characterization of the emitter; the suitable cavity is further fabricated for the emitter [Dou08]. In this approach, for nano-scale cavities, the fabrication needs to be realized in very close proximity to the emitter, with the risk of damaging the emitter properties and thus jeopardizing the performance of the device.

In DELICACY, we propose an alternative path to reach the spectral and spatial matching based on open cavities where one mirror and the field concentrator are fabricated on the apex of an optical fiber. The spectral and spatial matching are obtained by moving the fibered mirror relative to the flat mirror supporting the nano-emitters. Fibered Fabry Perot cavities are nowadays routinely used for a large range of solid-state emitters. They can be an interesting alternative to monolithic cavities for epitaxial quantum dots [Naj19], but have also proven to be very relevant for emitters that are not easily compatible with conventional clean room fabrications techniques such as single molecules [Wan19], 2D material [Sid19] or carbon nanotubes (including our contributions) [Jea16, He18]. Partner LPENS has been using Fibered Fabry-Perot (FFP) cavities for several years.

Due to the near-field nature of this sub-wavelength cavity, the field amplitude decreases on a 50 nm length scale (in the axial and lateral direction) away from the field concentrator. Hence, efficient coupling to the field can be reached for nano-emitters deposited at the surface of the back mirror rather than buried in a matrix. The DELICACY project mainly focuses on graphene quantum dots (GQDs) but other innovative processable materials could also be used. Over the last fifteen years, a great attention has been paid to the development of nanographene materials, in particular graphene quantum dots, using the conventional top-down approaches based on oxidation, sonication and thermal treatments of bulk graphitic materials [Sun08, Peng12, Li13]. However, the top-down approach does not permit to control the structure at the atomic level. In particular, it does not allow sufficient control of the morphology and oxidation state of the edges, which drastically impacts the properties. As an example, it was demonstrated that, at the single nanoparticle level, the PL arises from the edge defects and that the uncontrolled edges blur out the intrinsic properties of GQDs [Xu13, Lei20]. This example points toward the necessity of producing graphene materials with a level of control beyond the capabilities of top-down methods. To achieve atomically precise graphene materials, the bottom-up approach must be considered.

At the end of the 90's, the group of K. Müllen at the Max Planck Institute for Polymer Research in Mainz developed the chemical synthesis of large polycyclic aromatic hydrocarbons (PAHs) [Mor97, Wu07, Xu20]. These materials are similar to small pieces of graphene exhibiting atomically controlled shapes, edges and functionalities. While the photophysics of hexabenzocoronene (one of the smallest members of the GQD's family) is relatively well-known [Hug12, Hai21], only few optical experiments (absorption, photoluminescence spectroscopies and time-resolved experiments) have been reported on larger GQDs and a lot of work needs to be done to relate the spectra to the electronic structure of the objects. In 2018, the consortium reported for the first-time single molecule spectroscopy experiments on bottom-up $C_{96}C_{12}$ GQDs [Zhao18]. Despite the tendency of these $C_{96}C_{12}$ GQDs to form columnar aggregates [Tom04], we were able to individualize them in trichlorobenzene and in polystyrene matrix and we have shown that they are efficient and stable single-photon emitters at room and at cryogenic temperature [Zhao18, Liu22a, Liu22b].

c. Methodology and risk management

The main objective of the DELICACY project is to efficiently couple a single GQD to a high-quality factor, deeply sub-wavelength mode volume cavity. To reach this objective, the project is divided into three interdependent work packages.

The first part is dedicated to the fabrication and characterization of nano-cavities by partner LPENS. It includes different cavity geometries in order to go step by step in cavity complexity, starting from a planar cavity including a dielectric concentrator for fast characterization and reaching progressively more involved fibered cavity geometries to allow spatial and spectral matching with GQDs.

The second part regards the bottom up synthesis of graphene quantum dots and their optical properties. Building on the already available GQDs, the relation between structure and properties will be investigated, together with the quantum light properties of GQDs. This involves partners NIMBE, LUMIN and LPENS.

Finally, a third part is dedicated to the study of GQDs coupled to a nano-cavity. Even if this part can be started from the early stage of the project, it will benefit from the progress of the two other parts. Room temperature efficient single photon sources, ultra-sensitive absorption spectroscopy, and the physics of strong light-matter coupling will be investigated. Partners LPENS and LUMIN will be involved in this section.

WP0: Management of the project: (Task Leader Y. Chassagneux)

DELICACY will be coordinated by Y. Chassagneux in close interaction with the other partners. The WP0 is devoted to the organization of the project and to the management of the involved participants. A peculiar attention will be given to the progress of the project, with regard to the objectives and to data management. All partners have worked in close collaboration for more than a decade, no problem of communication is expected. Bi-annual meetings are planned (face to face, or via videoconference depending on the health situation). An internal wiki-type web-page will be maintained by the PI, for data sharing and reporting of the progress of the project.

WP1: Cavity fabrication and characterization (LPENS) (Task Leader Y. Chassagneux).

Some steps in the fabrication and characterization of the cavity will be done with the subcontractor *atom chips* group led by J. Reichel (LKB, Paris) who is one of the pioneers in Fiber Fabry Perot (FFP) cavity technology and with whom the LPENS partner has been developing the first generation FFP cavities. It includes fiber preparation and shaping by CO_2 laser ablation, and helping in cavity active stabilization. The subcontractor lab is physically in the same building as partner LPENS, allowing for very easy interaction.

Task 1.1: Geometry of the cavity (LPENS)

Cavities that will be used in DELICACY are open cavities based on miniature Fabry-Perot like structure. Conventional micro Fabry-Perot cavities have a mode volume limited by diffraction ($V \gtrsim \lambda^3$). The core of the DELICACY project is to integrate a dielectric field concentrator inside the Fabry-Perot structure in order to strongly reduce the mode volume.

The nanocavity will consist of a dielectric nanoantenna (the field "concentrator") placed on one of the two facing mirrors (see Fig 3.). While targeting dielectric mirrors to reach higher Q factors, we could also

use metallic mirrors in a preliminary step to facilitate and speed up the fabrication process. In fact, the high-quality dielectric reflector will be fabricated by an external company with an overall lead time of 2-3 months. To avoid any possible confusion, we stress that such a metallic layer purely acts as a mirror and is not related to any plasmonic effect.

To efficiently couple the GQD with the cavity, one of the two mirrors will be fabricated on the tip of the fiber. The fiber can be moved in the 3 directions in order to be able to tune the cavity around the GQD and to be able to match both spatially and spectrally to the emitter. Note that due to the field penetration in the dielectric mirrors, the smallest order Fabry Perot mode is reached for a vanishing air gap between the mirrors. Hence, a vertical length tuning of the cavity of the order of a few tens of nanometers is enough to reach the spectral matching with the emitter while remaining in the near field of the dielectric nano-antenna. For the metallic mirror approach a dielectric spacer will be added to the back mirror before depositing the nano-emitters. Partner LPENS has several years' experience [Jea16, Jea17, He18] in using fibered Fabry-Perot cavities (without the concentrator).

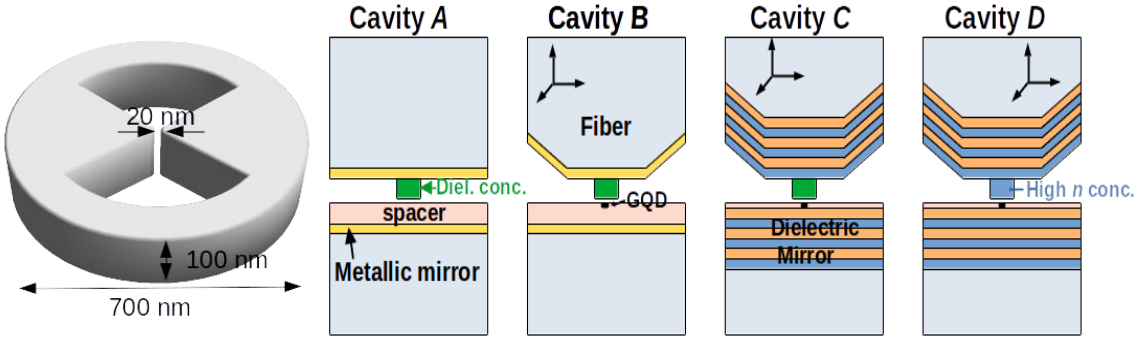


Fig. 3 : Left: concentrator geometry. Right: Cavities planned to be used during the project by order of increasing expected performance (and difficulty).

During the course of the project four types of cavities will be used by order of increasing expected performance (and difficulty). The first cavity (Cavity *A*, cf Fig. 3), is a planar cavity made of two facing metallic mirrors and including the dielectric "concentrator" made of e-beam resist. All the fabrication steps of this cavity can be performed in the clean room at LPENS and thus can be implemented in the very near future. With this planar configuration, several "concentrator" geometries will be fabricated on the same metallic mirror for fast throughput of cavity testing. This cavity *A* is not intended to be coupled to GQDs, thus no mechanical displacements are needed. This cavity will be mainly used to probe the "concentrator" optical properties and compare the results to numerical simulations in order to obtain feedback to the fabrication. The cavity *A* will be mainly used at the early stage of the project, but can be used later to eventually probe new designs of "concentrator".

The cavity *B* (cf Fig. 3), is an evolution of cavity *A* to become compatible with the coupling to GQDs. One side of the cavity will be fabricated on the sharpened tip of a fiber (metallic mirror and dielectric resist based "concentrator") to allow for precision positioning of the cavity on the GQD. This cavity will combine a small mode volume $V \sim 0.02\ \lambda^3$ but still with a moderate Q factor $Q \sim 100$ due to ohmic losses in the metallic reflecting layer.

The cavity *C* is a full dielectric cavity. The dielectric mirror will be deposited by an external company and consists of alternating layers of TiO_2 and SiO_2. The confined mode being subwavelength it will be coupled to external optical mode in all directions. Common $\lambda/4n$ dielectrics stacks, cannot be used as their high reflectivity are limited in the angular domain. Nevertheless, other geometries of the dielectric stack can be chosen to form an omnidirectional mirror [Fin98, Deo01] which will prevent high angle losses. Note that the in-plane confinement is ensured by the external annular structure of the concentrator, similarly to a micropillar geometry.

The cavity *D* is similar to cavity *C* but with a high optical index concentrator (made of TiO_2) to further reduce the mode volume.

Deliverable:

D1.1a: t_0 +2 Design of cavity A & B

D1.1b: t_0+12: Design of cavity C & D

Task 1.2 Fabrication (LPENS)

Fiber sharpening:

With our fibered cavities setup, the fiber typically presents a small angle (few degrees) with respect to the normal axis of the flat mirror. This small angle prevents small cavity mode volume to be reached as the edge of the fiber touches the planar mirror. To solve this issue fiber will be firstly sharpened in a pencil like shape. This step will be done by CO_2 laser ablation, already mastered by partner LPENS in close collaboration with subcontractor J. Reichel group (LKB). The risk related to CO_2 laser ablation of fiber is very low as this technique is successfully routinely used by partner LPENS (cf Fig. 4).

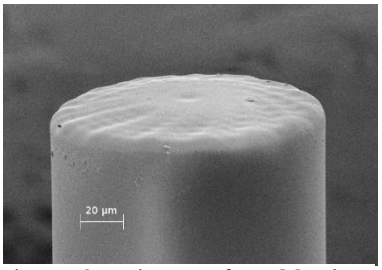


Fig 4: Sem image of a CO_2 laser micromachined fiber done by partner LPENS.

Mirror coating:

During the project two types of mirrors will be used: either metallic mirrors or dielectric mirrors. Metallic mirrors will be deposited in the LPENS clean room and the dielectric coating will be made by an external company (LaserOptik) by Atomic Layer Deposition leading to high quality dielectric structures. Metallic mirrors will be mainly used in the beginning of the project. Their relative high losses are counterbalanced by the fact that they can be produced very fast. Dielectric mirrors will be used in a second step to reach high quality factors.

"Concentrator" fabrication:

Concentrator will be made by e-beam lithography and in a first step, the resist itself will be used as optical material. Specifically, the negative resist HSQ (Hydrogen silsesquioxane) will be used. After exposition and annealing, this resist transforms into silica. For dielectric antennas, both a nano-gap antenna and a nano-bridge antenna lead to the E-field anomaly used to generate the strongly sub-diffraction mode volume. The E-field anomaly strength directly scales with the inverse of size of the nano-gap or nano-bridge at the surface of the concentrator. A sharp vertical edge of the resist structure is not necessary to reach very small mode volumes, reducing the risks and constraints on fabrication. Note that oblique resist edges are even better as they make an adiabatic passage between the flat reflecting layer and the concentrator surface.

E-beam lithography on fiber:

To perform clean-room processing on the tip of a fiber, the fiber cannot be used directly as its surface is too small to deposit the resist homogeneously. The fiber will be inserted in a larger holder. In order to obtain a correct resist deposition, the tip of the fiber should be flushed with the top of the holder. To precisely fix the fiber in the proper vertical position, optical imaging or using a Miro interferometry imaging with sub-micron vertical resolution will be used to monitor the fiber position. For e-beam lithography the same holder will be used to maintain the fiber position during the lithography. Clean room fabrication on a fiber is not a common procedure, nevertheless it is well referenced in the literature, see for example [Kos14].

High step index concentrator

The exponent of the E-field divergency scales like the dielectric constant of the concentrator minus 1 (air index). To further reduce the mode volume using a high index material would be valuable. To get a high step index concentrator we are planning to fabricate a TiO_2 dielectric structure (n=2.55 for λ=700 nm). To do so, the dielectric Bragg layer (made of a SiO_2 - TiO_2 stack) will include an upper TiO_2 layer that will be etched in the desired geometry [Hay14].

Deliverable:

D1.2a: t_0: concentrator fabrication on a glass substrate

D1.2b: t_0+6: e-beam lithography on fiber

D1.2c: t_0+18: Tunable low V high Q cavity

D1.2d: t_0+24: High index concentrator

Risk: Difficulties with e-beam lithography on fiber: likelihood low, impact moderate.

Fallback solution: The concentrators can be fabricated on the flat mirror, and the GQDs deposit on the fiber. In that case the fibered optical port may not be used as the emitter will not be positioned in the axis of the fiber core. Nevertheless, the optical access from the flat mirror can still be used.

Risk: Etching of TiO_2 leads to large rugosity. Likelihood: moderate, impact low.

Fallback solution: The fabrication of the TiO_2 concentrator can be done without etching the TiO_2, but with a lift off technique. It involves a deposition of TiO_2 after the e-beam production of the inverse geometry of the concentrator, which after the lift off gives the desired structure. This technique has been used to demonstrate high Q (Q>150 000) ring resonators [Qiu15].

Task 1.3 Mechanical vibrations control (LPENS)

Open cavities are very sensitive to axial vibrations. For a cavity with $Q \sim 10^4$, the vibration amplitude should be less than ~0.1 nm RMS. To reach these values, we will combine passive vibration suppression together with active feedback vibration suppression. Room temperature and cryogenic temperature experiments will require separate passive vibration suppression approaches due to various origin of the vibrations and to the strong variations of elastic properties of materials with temperature in this range.

At room temperature, vibrations of the cavity originate from the ones transmitted through the optical table and also from acoustic noise (and air flow) transmitted through the air environment. To strongly reduce these two components, we intend to place the cavity inside a vacuum chamber, itself on top of a commercial negative stiffness passive isolation platform with very low resonant frequency ~0.5 Hz. The quality of this passive isolation will be monitored directly by using a fibered Fabry-Perot cavity transmission or by using precise accelerometers. On the cavity side, effort will be made to increase the cavity rigidity and thus increase as much as possible the cavity mechanical resonant frequency.

At cryogenic temperatures, we will use a closed-cycle cryostat (which allows us to continuously work on the same emitter for months). The drawback of such a system is the vibrations generated by the helium compressor. We presently have a *Montana Instrument* closed cycle cryostat. Despite the ~1 Hz compressor hammer-like noise, the cold stage shows a 5nm peak to peak vibrations amplitude only. This system is already implemented by partner LPENS with fibered cavities (without concentrator). Using appropriate passive vibration suppression design we routinely obtain a cavity length vibration amplitude of ~1nm rms.

Both at room temperature and low temperature, to further suppress vibrations, an active stabilization technique will be employed with a Pound Drever Hall method [Dre83]. A resonant modulated laser is sent to the cavity; after demodulation of the transmitted beam, the signal is directly proportional to the cavity length. This signal is reinjected (via a PID loop) in a fast-piezoelectric actuator to lock the cavity length. In order to avoid dazzling the GQD PL emission with the stabilization laser beam, higher lateral order cavity modes can be used for stabilization purposes.

Deliverable:
D1.3a: t_0+9: Low vibration platform
D1.3b: t_0+15: Pound Drever Hall stabilization of cavities.
Risk: Difficulties to reduce vibrations. likelihood: low, impact: medium
Fallback solution: Another approach to get rid of the detrimental effect of vibrations, is to introduce controlled sinusoidal oscillations of the cavity length and to extract the cavity position from spectral measurements. Subsequent data treatment allows retrieving the vibration-free PL properties. For cryogenic experiments the main vibrations source is the Helium compressor, with a ~1Hz hammer blow like noise. Hence, measurement could also be synchronized with the compressor, with a boxcar-like acquisition, recording data only on limited time windows between the compressor kinks.

Task 1.4 Optical characterization of cavities (LPENS)

Cavities will be probed in transmission to deduce the quality factor and the tuning range. By modulating the cavity length faster than the vibrations, the deduced cavity parameters are not impacted by detrimental vibrations.

The concentrator supports two orthogonally polarized modes, one contains the E-field anomaly leading to the sub-diffraction mode volume, and the other one has a more classical shape. These two modes are not degenerate, and can be identified by investigating the polarization response of the cavity. Comparison between the “normal” and the “anomalous” mode will give insight into the cavity properties, in particular to the effect of strongly confined light to the quality factor.

The far field pattern of the cavity will also be probed by measuring the back focal plane images (in transmission) to evaluate precisely the angular response of the sub-diffraction cavity and check if it is

compatible with the expected near-field. The cavity mode volume cannot be accessed directly by probing empty cavities. To experimentally measure the near field pattern, from which the mode volume can be directly computed, we will deposit gold nanoparticles on the flat mirror. They will be used as nm local probes revealed by extinction measurement of the cavity transmission. Gold nanoparticles are easily available commercially and are well documented. As an example, gold nanorods of 20x10 nm exhibit a very broad resonance ($\delta\lambda \sim 50$ nm) with a large extinction cross section $\sigma \sim 10^4$ nm^2 [Cru14]. This value corresponds to a few percent reduction of transmission for a single pass focused beam, which would correspond to a very large reduction of transmission in cavity. The near field map is obtained by scanning laterally the fiber with the "concentrator" respect to a fixed position of the nanorod, the accuracy being given by the particle size.

Deliverable:
D1.4a: t_0+9: "Concentrator" optical properties.
D1.4b: t_0+21: low V high Q cavity optical properties.

Task 1.5: Numerical simulations (LPENS)

Finite element numerical simulations will be performed to refine the design of the cavities and to compare to the measurements. Here the simulation difficulty comes from the different length scales in the cavity: the "concentrator" contains a tiny nm scale structure, embedded in a dielectric reflector of several micrometers. For the mode volume calculation, the dielectric reflector can be simplified to a few pairs, reducing the computational resources needed. 3D simulations can also be compared to much simpler 2D simulations of the "concentrator" plane to get an empirical relation between the 3D mode volume V_{eff} and to the 2D effective surface S_{eff} which can be computed easily. Preliminary simulations show that $V_{eff} \sim 0.66\, S_{eff}\, \lambda$ (cf Fig.5).

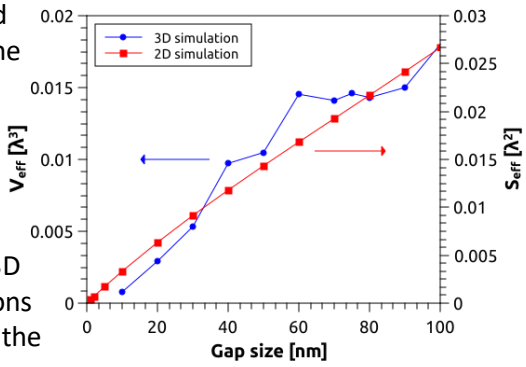


Fig 5. 3D (resp 2D) simulation of the mode volume (resp. surface) as a function of the gap size.

The second crucial parameter of the cavity is the quality factor. Here the structure cannot be approximated by a 2D counterpart and the actual structure would require too much computational resources to be properly simulated. Nevertheless, since the cavity is essentially a layered structure, other numerical strategies can be used: One can rely on mode matching technique to decompose the structure in 2D planes with a complete basis that are matched at interfaces. Note that mirrors have a transverse translational symmetry and can be efficiently computed in the Fourier domain. This mode matching strategy is already implemented at LPENS, and has been verified with a simpler structure, *i.e.* a disk instead of the concentrator, where 3D simulations are possible (by using the axial symmetry).

Deliverable:
D1.5a: t_0: Simulation of low V medium Q cavity already available (cavity *A* & *B* Fig.3).
D1.5b: t_0 + 6: Simulations of low V high Q cavity (cavity *C* Fig.3).
D1.5c: t_0 + 12: Simulations of low V high Q cavity with high step index (cavity *D* Fig.3).

WP2: Synthesis and optical characterization of GQDs (NIMBE LUMIN LPENS) (Task leader S. Campidelli).

Task 2.1 Chemical synthesis of GQDs (NIMBE)

The GQDs used in DELICACY will be synthesized following the bottom-up approach (organic synthesis method) which permits the access to fully controlled structures [Xu20, Nar15]. Partner CEA-NIMBE has already synthesized several GQDs (Fig. 6a), among them the C_{96}-$(C_{12}H_{25})_6$ [Zha18] and C_{96}-$(t\text{Bu})_6$ therefore, they will be available at t_0 for the consortium. New structures will be developed during the project; in particular, we will synthesize rod-shaped (Fig. 6b) and square-shaped (Fig. 6c) GQDs. These particles are interesting for several reasons:

- Their topology is very different from those of Fig. 6a giving rise to different bright and dark states and so different optical properties. For example, the two GQD C_{96}-(tBu)$_6$ (Fig. 6a) and C_{96}-(tBu)$_8$ (Fig. 6b) contain the same number of sp^2 carbon atoms (96) but their shape/edges are different; we expect very different physical properties for these compounds.
- The family of GQDs described in Fig. 6b exhibits the same topology, but they differ by their length (*i.e.* by the number of terphenyl moieties (in red) in the molecules). They constitute an ideal starting point to study the structure-properties relationship and to tune the emission wavelength toward near-infrared.

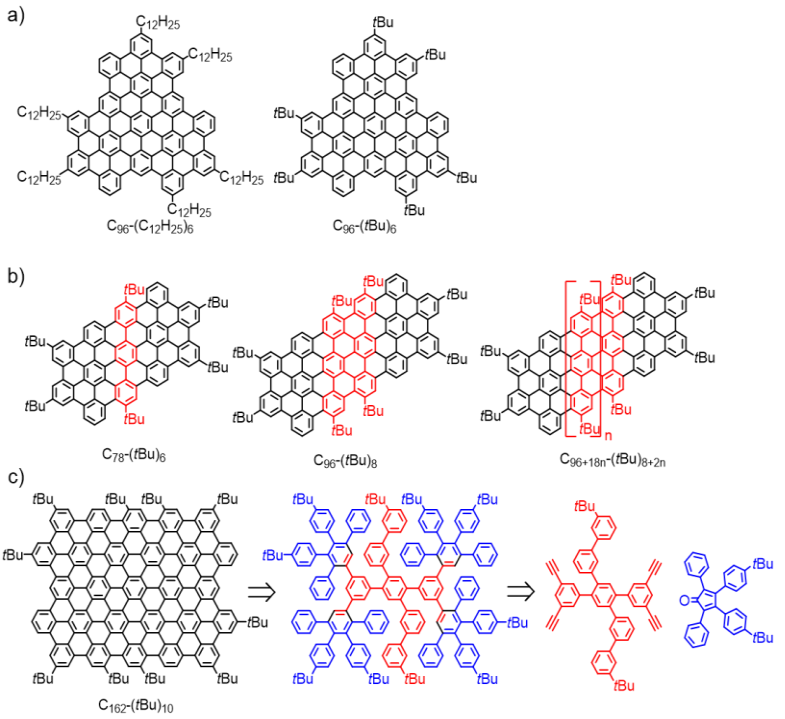


Fig. 6. Example of GQDs that will be studied. a) Triangular shaped C_{96} GQDs containing different solubilizing groups (already available at t_0) b) rod-shaped GQDs containing tert-butyl solubilizing groups, we will produce similar GQDs in terms of shape with different length to study the structure property relationship; c) example of larger GQD that will be produce for the project and retrosynthetic scheme for its synthesis.

The exact GQDs structures can be achieved thanks to well-established and mastered organic synthetic methodologies. The synthesis of all these GQDs is based on the same concept that is exemplified for C_{162}-(tBu)$_{10}$ (Fig 6c). First, a core containing acetylenic functions will be synthesized (molecule in red – Fig 6c); it will react with cyclopentadienones (molecule in blue – Fig 6c) to give the polyphenylene structure that will be oxidized via the Scholl reaction to give rise to the corresponding GQDs. The functional groups at the periphery (especially tBu groups) will ensure a good solubility of the GQDs in organic solvent, despite their large sp^2 structures. These groups permit to individualize GQD in solution, it will ensure their good dispersion as single objects in the cavity.

Finally, other photoactive molecules emitting in the red or infrared can be envisioned during the project. We recently reported the synthesis of fused a mono-anthracene porphyrin derivative [Pij21], this pi-conjugated porphyrin exhibits an emission maximum at 880 nm. We are currently working on the synthesis of fused bis- and tetra-anthracenyl porphyrins in collaboration with EMPA (in Switzerland) and Swansea University (Wales)[Bal22]. DELICACY could benefit of these new photoactive molecules.

During the duration of the project, constant discussion with the consortium will be pursued in order to design and synthesize new potentially interesting structures towards improving light-matter coupling in the cavities.

Deliverable:
D2.1a: t_0 – triangular shaped C_{96} GQDs available
D2.1b: t_0+9 – synthesis of rod-shaped GQDs
D2.1c: t_0+24 – synthesis of larger GQDs
D2.1d: t_0+36 – design and synthesis of other structures according to discussion with the consortium

Risk: Problem with the synthesis of GQDs; likelihood low, impact medium.
Fallback solution: As mentioned above, several GQDs are already available at t_0 for the consortium. The synthesis of the GQD precursors is based on classical organic reactions well mastered by Partner NIMBE; it does not present any risk. The last step consisting in the oxidation of the polyphenylene dendrimers with $FeCl_3$ to form the GQDs constitutes the key reaction; if the reaction is not properly controlled, it can bring to partially chlorinated GQDs. If so, other oxidative reagents will be tested, in particular 2,3-dichloro-5,6-dicyano-1,4-benzoquinone (DDQ) in the presence of triflic acid which is actually very popular for the

formation of polycyclic aromatic hydrocarbons (PAH). We will also take advantage of our strong collaboration with the group of K. Müllen (pioneer on this kind of chemistry).

Task 2.2 GQD matrix (NIMBE, LUMIN)

An essential challenge for efficient GQD-cavity studies is developing a sample with stable, isolated GQDs of well-controlled intrinsic optical properties. This task aims to determine the best host matrix for the GQDs to use inside the cavity.

As a first step, we will use the know-how we developed in the study of C_{96}-$(C_{12}H_{25})_6$ GQD by partners LUMIN and LPENS [Zha18, Liu22a]. First, GQDs will be individualized in solution and then dispersed in a polystyrene matrix by spin-coating on top of the flat-mirror part of the cavity. Here, the thickness is typically of the order of one hundred nanometers.

Polystyrene matrices allow the study of stable single GQDs, with reasonably narrow linewidths (sub meV at ~10K) [Liu22b]. However, the linewidth remains far from the lifetime limit, which is needed for advanced cavity quantum electrodynamical effects. Therefore, to improve the quality of the GQD emission and reduce the sample thickness toward better coupling to the cavity field at the GQD position, we envision the use of alternative matrices.

A first approach will be to co-crystallize by precipitation the GQD of interest with non-active molecules like *para*-terphenyl or anthracene [Paz18]. With this technique, small nanocrystals of tens of nm can be obtained. Note that anthracene solution containing the molecule of interest can also be spin-coated to form crystalline films of a few tens of nm [Ton10].

A second approach will be to use Shpol'skii matrix (typically aliphatic alkanes like heptane, octane, or longer) to disperse GQDs. Such matrices have been previously used to obtain Fourier transform-limited linewidth on single molecules' emission [Pot11]. We expect that the GQDs will have a relatively low solubility in alkanes; co-solvent like toluene or tetrahydrofuran could be necessary to disperse the molecules.

For the different matrices, first, we will characterize the matrix thickness with an AFM or a profilometer. Then, we will study the sample with deposited GQD on a micro-photoluminescence setup, both at room and cryogenic temperatures. We will test the dispersion and the stability of the GQDs. Spectroscopy of single GQDs at low temperature will provide a first estimate of their emission linewidth [Liu22b]. Additionally, other photophysical properties of the GQD in the considered matrix, such as spectral jitter, lifetime and intersystem-crossing, give important information on the intrinsic nature of GQD emission and the coupling to its environment.

Deliverable:
D2.2a: t_0 +6 – Sample of C_{96} GQD in different matrices on dielectric mirrors
D2.2b: t_0+18 – Sample of GQDs in anthracene nanocrystals
D2.2c: t_0+18 – Sample of rod-shape GQD in different matrices on a dielectric mirror

Risk: The GQD are not fitting in either of the alternative matrices. likelihood low, impact low.
Fallback solution: The polystyrene matrix that we demonstrated for C_{96} GQD is sufficient for most of the targeted objectives of the project, even if some figures may be lower than expected.

Risk: difficulty in producing thinner matrices hosting GQDs. likelihood medium, impact low
Fallback solution: Statistically, some GQD should be near the matrix interface. At the expense of experimental investigation, we could find some GQDs close enough from the surface that can be importantly coupled to the near field of the field concentrator.

Task 2.3 Optical Characterization of GQDs (LPENS, NIMBE, LUMIN)

The objective of the optical characterization task will be two-fold. First, in conjunction with task 2.2, it will allow characterizing the effect of the matrix onto the GQD photophysics. Then, it will focus on the GQDs' structure-properties relationship, taking advantage of the various shapes and sizes available from the chemistry and provided through task 2.1. Beyond the fundamental interest of understanding the structure-properties relationship, this step will allow choosing the appropriate GQDs for the cavity experiments and to guide chemistry toward the fabrication of GQD structures of physical interest, for example, increase the size of the object to red-shift its emission or increase the number or the size of the solubilizing groups to decrease the stacking and improve the dispersion in matrices.

To address the structure-properties relationship, we will first use ensemble optical experiments. Photoluminescence and ultrafast spectroscopy will thus provide crucial information on the GQDs emission wavelength, electronic structures, and quantum efficiency. Therefore, it will allow fast screening of the available GQDs' photophysical properties, potentially selecting the most appropriate for the cavity experiment.

Using single-molecule spectroscopy, we will then address the intrinsic properties of individual GQDs, thus avoiding inhomogeneous broadening. By carrying out experiments at low temperature (5K – 300K), we will study the evolution of the luminescence spectrum of single GQDs with temperature, providing insight into their vibrational modes. These studies will first allow characterizing the effect of the matrix but also identifying the GQDs vibrational modes. Describing these modes is of fundamental importance since they strongly impact the photophysics of GQDs, for example, brightening dark transitions and hence the overall luminescence quantum yield of the GQD [Liu22b].

Additionally, low-temperature spectroscopy of single GQDs will provide seminal information on the emission linewidth and spectral jitter of the considered GQD. Besides using time-resolved luminescence, we will study the dynamics of photon emission, providing important insight on the existence of shelving states and intersystem crossing that are expected to impact the light emission efficiency.

Our optical characterization results will be discussed in light of the theory work of David Beljonne's group (University of Mons) through an ongoing collaboration that included the three partners of DELICACY (Flag-ERA "OPERA"). The continuous feedback between experiment and theory will be an asset to understand the structure-properties relationship deeply and guide chemical synthesis.

Deliverable:
D2.3a: t_0+12 – Characterization of matrices using single C_{96} spectroscopy.
D2.3b: t_0+18 – Spectroscopic characterization of rod-shape GQDs. Highlight on structure-properties relationship.
D2.3c: t_0+36 – Spectroscopic characterization of larger GQDs.

Risk: This task depends on spectroscopic tools available by the consortium. The main risk is related to the GQD fabrication task2.1. likelihood low, impact medium.
Fallback solution: As in the taks2.1, if some GQD are not available we will rely on other GQD, rather already available, or through external collaborations.

Task 2.4: Quantum light properties (LPENS, LUMIN)

An important aspect of the project is the opportunity to take advantage of GQDs as single photons sources, for quantum technologies. It is thus essential to address the quantum properties of emitted photons, such as their purity, indistinguishability and coherence. In the context, second order correlation function measurements will be performed by means of a Hanbury-Brown and Twiss setup. First, it will provide significant insights on the purity of the emitted photons. Then it will complement the spectroscopic results of task 2.3, by probing the internal conversion yield of different GQD geometries. In fact, it was recently recognized that the luminescence quantum yield of GQDs is related to the intersystem crossing.

In a second time, the indistinguishability of emitted photons will be addressed by Hong-Ou-Mandel measurements performed at low temperature. From the indiscernibility figure of merit, the dephasing time T_2 of GQDs will be extracted. This value cannot be probed by spectral measurements as spectral wandering at short time scales hide the relation between linewidth and dephasing time T_2.

Deliverable:
D2.4a: t_0 – Single photon purity, lifetime and intersystem crossing of C_{96} GQDs geometry
D2.4b: t_0+18 – Indistinguishability figure of merit of C_{96} GQDs
D2.4c: t_0+24 – Effect of the matrix environment for the quantum light properties of GQDs
D2.4d:t0+36 - quantum light properties of larger GQDs

Risk: The quantum light properties of the GQDs are not as good as expected. likelihood low, impact low.
Fallback solutions: Other GQD may be investigated. Even if quantum properties of GQDs are not at the state of the art of single emitters, most of the project can be done, and applications to the study of strong coupling and absorption spectroscopy remains valid.

WP3: GQDs in dielectric nano-cavities. (LUMIN LPENS) (Task leader C. Voisin).

Task 3.1: Spectral and spatial matching (LPENS)

Open cavities offer the unique opportunity to compare a nano-emitter in free-space and in a cavity, with exactly the same environment (temperature, local electrostatic bath, excitation conditions…) as recently demonstrated by partner LPENS [Jea16, He18] with fiber cavities (with no concentrator). The accuracy of the positioning of the emitter is obtained by fine piezoelectric motion, combined with optical control leading to a sub µm position error. With the cavities proposed in this project (cavity *B*, *C* or *D* of Fig 3) even if the GQD is laterally misplaced with respect to the field concentrator, the vacuum Rabi coupling *g* is already larger than 50 µeV, resulting in a sizeable brightening of the emitter, and can then be optimized by scanning laterally the fiber (and thus reaching the field concentrator) to reach a much larger coupling up to *g* ~ meV in our simulations (cf Fig. 7).

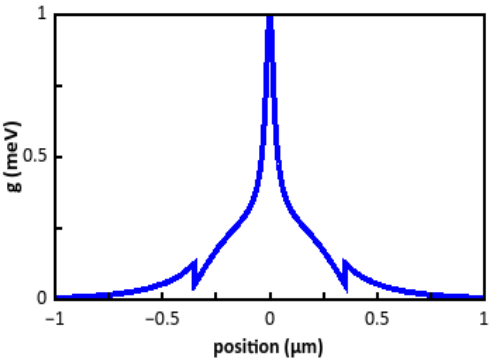


Fig7. Vacuum Rabi coupling g in function of the distance between the emitter and the concentrator at the center of the fiber tip. The discontinuity at 350 nm corresponds to the external edges of the concentrator.

Deliverable:
D3.1a t0+9: protocol for spatial matching of the emitter to the cavity

Risk: Damaging the concentrator when laterally moving one mirror. Risk: medium, impact: very low
Fallback solution: Moving laterally can be done safely in 3 steps: a) enlarge the cavity of few microns. b) move laterally. c) close again the cavity. This procedure is already implemented by partner LPENS

Task 3.2: GQD in nano-cavity with metallic reflecting layer (LPENS, LUMIN)

GQDs will be firstly coupled to the low-V, moderate Q cavity (cavity *B* in Fig. 3) at room temperature. In this configuration the Purcell enhancement factor will be investigated in these innovative cavities. The Purcell effect and the vacuum Rabi splitting will be deduced from the measurement of the acceleration of the PL decay rate of the GQD inside the cavity compared to the one in free-space. This will be done by using the expertise of partner LPENS which has developed an experimental setup to measure the very same emitter, with exactly the same environment in free space and in cavity. Note that this metallic mirror configuration is perfectly suited for room temperature application where phonon broadening leads to broad PL line widths of the emitters and a sizable Purcell brightening is expected even at room temperature.

"Concentrators" that will be used in DELICACY supports two cross polarized modes: the first one has an E-field oriented in the direction of the dielectric part of the concentrator. This is the mode with a diverging E-field in the center, thus responsible for ultra-low mode volume. The second mode, does not present any peculiarity, and correspond to mode volume in the order of λ^3. The comparison between the "normal" mode and the "anomalous" one will be exploited to get a further insight in the effect of the "concentrator". Elongated GQDs with a highly polarized optical response could be used to probe this effect.

Deliverable:
D3.2a: t_0 +12: Single GQD coupled to nano-cavities
D3.2b: t_0 +15: Purcell acceleration
D3.2c: t_0 +15: Comparison of normal mode and anomalous mode coupling

Task 3.3: Room temperature tunable single photon source with a dielectric nanocavity (LPENS)

At room temperature, solid state single photon emitters typically have a linewidth of ~10 meV due to phonon broadening. For typical diffraction limited dielectric cavities, the line is far too broad to yield any significant coupling to the cavity mode. The situation is different with a nano-cavity, for which the reduced coupling due to line broadening can be compensated by the ultra-small mode volume and can yield to an efficient coupling to the cavity mode even at room temperature. In this configuration, the cavity "forces" the GQD to emit photons with cavity energy and linewidth, and an efficiency (fraction of photons emitted in the cavity mode per excitation pulse) higher than 90% is expected. Using a small axial displacement, the tuning

range of the proposed single photon source can represent up to several thousand times the spectral width of the cavity, opening the way to wide-range multiplexing in quantum communications.

The first important parameter for a single photon source is the purity of the emission, i.e. the probability of emitting spurious pairs of photons should remain very low. This aspect is probed by the value at zero delay of the second order correlation in a Hanbury-Brown and Twiss setup. For C_{96} GQDs a very good single photon purity has been measured by the consortium without cavity: $g^2(0) < 0.1$ [Zha18]. When coupled to the cavity the single photon purity is usually improved since the cavity filters out spurious emission which can originate from defects in the surrounding matrices and other parasitic light.

In addition to the single photon purity, the indistinguishability of emitted photons is an important feature of single photon sources, which is measured using a Hong-Ou-Mandel setup. Even if indistinguishability is not required for simple quantum telecommunication protocols, it becomes necessary for more involved devices, such as in quantum repeaters or for quantum computing protocols. In a simplified picture, the indistinguishability is obtained when the spectral linewidth of the photon is limited only by the emitter lifetime. At room temperature, due to thermal broadening, no solid-state emitters can produce directly indistinguishable photons (linewidth between 3 to 6 orders of magnitude larger than intrinsic linewidth deduced from lifetime [Gra19]). When coupled to a cavity, the situation is different: with the Purcell effect, the emitter emits preferentially in the narrow cavity mode (due to the so-called cavity feeding effect [Auf10]); the ratio of the emission rate in the cavity mode to that of other modes is in fact given by the Purcell factor. The emitter – cavity system thus acts as an effective composite emitter, with an effective linewidth given by the cavity one. Engineering of the optical cavity is indeed a powerful tool that can lead to the emission of indistinguishable photons, provided that a high enough quality factor is reached (cavity linewidth on the order of the inverse emitter lifetime). Beyond this simple picture, the indistinguishability has been explored theoretically both for conventional λ^3 cavities [Gra19], and deep subwavelength dielectric cavities [Cho19]. Two figures of merit describe the single photon emission: the degree of indistinguishability and the brightness (overall single photon emission efficiency subsequent to an excitation pulse).

For a cavity with $V_{eff}=0.01\ \lambda^3$, the vacuum Rabi coupling *g* should be ~1 meV. For a single GQD, the room temperature linewidth is $80\ \mathrm{meV}$, with a lifetime of $5\ \mathrm{ns}$ [Zha18] (note that other emitters have similar values at room temperature). For a cavity with a $Q=10\ 000$, and a coupling $g=1\ \mathrm{meV}$, the efficiency is 99.7% together with a $14\ \%$ indistinguishability [Gra15]. The expected photon flux is very high and should reach 6.10^{10} photons/s at saturation. With the nanocavity the product of efficiency times the degree of indistinguishability is at least a factor of 5000 above that obtained by simple spectral filtering of the bare emitter. For applications where the indistinguishability of photons is strongly required, the high photon rate allows to lose photons by using classical spectral filtering to reach a near unity indistinguishability figure of merit. Even with this reduction of efficiency, nearly indistinguishable photons, in a single mode fiber, with a flux larger than 10^9 photons/s can be achieved. This efficient, room temperature, single photon source is additionally tunable (by scanning the cavity). Note that photons characteristics can be further improved by using cascaded cavities [Cho19].

Deliverable

D3.3a: t_0 +36: Room temperature, tunable, efficient single photon source

Risk: Degradation of the performance due to vibrations: Risk: medium, impact: low

Fallback Solution: Lateral vibrations are estimated to be ~1 nm rms, this value is much lower than the shortest length scale of variations of fields. Thus, it won't impact the coupling. The axial vibrations can be more problematic as they modify (slowly in respect to other time scales) the cavity resonant frequency and can blur the observations. Due to the high photon flux expected, measurements can be performed faster than vibrations. Additionally, at room temperature vibrations can be greatly reduced by improving passive isolation with sorbothane foams or other elastic materials.

Risk: Photons exiting the cavity have a bad spatial shape: Risk medium, impact medium.

Fallback solution: The cavity has two exit ports: the fiber side or the planar side. The exit port can be chosen by using asymmetrical reflectivity between the two sides. The single mode fiber port is preferred as photons are directly funnel into the easy-to-use single mode fiber. However, the mode mismatch between the fiber mode and the cavity mode can lead to extra insertion losses. This point can be improved by replacing the

fiber with a coreless fiber (i.e. a homogeneous glass cylinder) spliced together with a graded index fiber that realizes the desired mode matching. Another possibility is to use the planar port where classical optical techniques can be used to maximize the collection.

Risk: The Purcell factor is too strong and the limiting transit time is not any more the emission lifetime but the time decay from higher excited states used for non-resonant pumping. In that case the expected photon flux at saturation will be lower. Risk low, impact low.
Fallback solution: The decay time from non-resonant excited state to the upper state of the transition is not known but it is expected to be very fast, in the ps scale, thus faster than the Purcell accelerated lifetime.

Task 3.4 Absorption spectroscopy (LPENS, LUMIN)

In addition to probing the emission properties of GQDs, the high-Q low-V cavity can also be used as an original tool to perform advanced absorption spectroscopy. PL spectra of GQDs is dominated by the emission of the zero-phonon line, but it also contains secondary resonances related to phonon coupling and also unidentified peaks suggesting a more complex electronic structure. While PL emission originates mainly from the lowest lying states due to internal relaxation, absorption spectroscopy brings a more direct access into the electronic density of states. Comparison of these absorption transitions to DFT numerical simulation (through an ongoing collaboration outside of the present project) would deeply improve the knowledge about GQDs.

The cavity setup developed for this project provides an original and efficient way to perform such an investigation: firstly, the effective interaction cross-section is increased by the number of round trips inside the cavity and secondly the sub-diffraction mode volume leads to an effective beam cross section much smaller than the one obtained by tightly focusing a laser beam in free space. The overall improvement of absorption sensitivity, in comparison to free-space absorption, is given by $4/(1\text{-}R)\ S_{diffration}/S_{cavity}$ [Mad15], where R is the mirror reflectivity, $S_{diffraction} \sim \pi(\lambda/2)^2$ is the surface of a focused beam at the diffraction limit, $S_{cavity} \sim 1.5V_{eff}/\lambda$ (cf Fig.5) is the effective surface of the cavity mode. Absorption sensitivity can be improved by 6 orders of magnitude, passing from a reduction of transmission of 10^{-7} -10^{-6} for a tightly focused laser beam on the GQD to a sizeable drop of the cavity transmission.

Thus, we will be able to perform high sensitivity single GQD absorption spectroscopy by monitoring the transmission of the cavity when scanning its length. These measurements can be done either by injecting a white light beam into the fiber or by scanning a laser line matched to the cavity mode. This technique was applied with conventional fibered cavities to image individual gold nanoparticles and to measure their extinction cross-section [Mad15].

Deliverable:
D3.4a: t_0 +33: Absorption spectroscopy of single GQD

Risk: Intensity fluctuation blur absorption signal. likelihood: medium, impact: low
Fallback solution: In case the fluctuations of the light source would still be too large to detect the transmission changes, we could implement the cavity ring down spectroscopy technique, which relies on the reduction of the cavity life-time when an absorbing unit is introduced in the cavity [Ber00]. This technique is immune to the source fluctuations.

Task 3.5 Strong coupling regime (LPENS, LUMIN)

When cooling down the GQD, the strong coupling regime can be reached: for a cavity volume $V \sim 0.01\ \lambda^3$, a Rabi coupling g of $\sim 1\text{meV}$ is expected. For C_{96} GQDs we measured a linewidth of 1.6 meV at 10K [Liu22b], but there is room for improvement putting the strong coupling at reach. The linewidth of single GQD at cryogenic temperature is far from being well known and optimized, the value reported by the consortium (published in 2022) is barely the unique value in literature. As a comparison, carbon nanotubes (made of sp^2 carbon as GQDs) have seen a 50-fold reduction of the low temperature linewidth down to *20µeV* [Luo17]. Similarly, in the other extreme, small molecules as DBT or DBATT (also made of sp^2 carbon) exhibit a lifetime limited transition when the proper matrix is used [Pot11]. Once the strong coupling is reached, we will focus on the few photons non-linearity towards quantum information processing. A two-level system strongly coupled to a cavity is well described by the Jaynes Cummings (J.C.) model. The eigenstates of the cavity-emitter system are mixed light matter quasi-particles, called cavity polaritons. At resonance, the

nonlinear energy splitting of the polariton branches is given by $2\sqrt{n}g$, where *g* is the Rabi coupling, and *n* the number of photons in the cavity. These states can be observed by measuring the transmission of the cavity, which exhibits two peaks of transmission at the polariton energies. Due to this nonlinearity, the transmission of the cavity depends on the number of photons even at the smallest photon number. As an example, the transmission of the cavity at the empty cavity frequency minus g is possible only when the number of photons in the cavity is exactly one; hence the light exiting the cavity will be antibunched. This is the photon blockade, which transforms a classical light beam into a quantum one. A quantum phase gate can also be generated in the same geometry. These are a few of the many applications of strong coupling that could be developed in a follow up of this project, with GQDs or with other emitters.

Deliverable:
D3.5a: t_0 +42: Strong coupling regime
D3.5b: t_0 +48: Photon blockade

Risk: During the project the GQDs linewidth remains too large to enter the strong coupling regime. likelihood medium, impact medium.
Fallback solution: other emitters could possibly be used, as defects in 2D materials, single molecule, or carbon nanotubes.

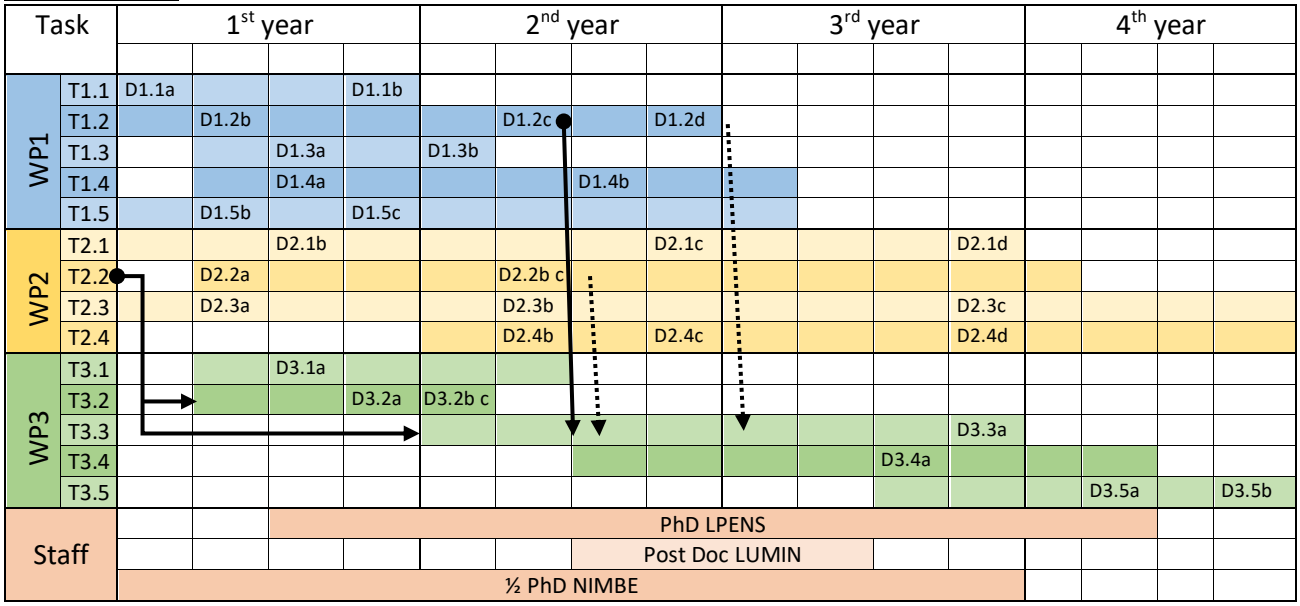
Risk related to health crisis or another crisis:

This project should not be too much impacted by the world situation (health or other), as it is mainly related to close-by fabrication and characterization facilities. Our institutions are also nowadays well organized to manage problems related to Covid pandemic crisis. We will anticipate as much as possible the purchase of equipment (small or large) which could be impacted by extended deadlines (electronic components, etc.) due to the health crisis, or other crises that may occur.

Gender-related aspects

Although our research activity has no connection with gender issues, it is clear that parity is far from being reached in physics labs in general. To improve the situation, even if the problem is much larger than our lever arm, we will promote gender equality during the project. During recruitment of PhD students or post-docs, for similar competence, our choice will go toward women. Additionally, during interaction with students of diverse degrees, we will encourage women promotion. In addition, our host institutions are very strongly involved in preventing all forms of gender inequality.

GANTT CHART



As can be observed in the previous chart, WP1 and WP2 are parallelizable, reducing the risk of a bottleneck in the development process. We stress here again that the availability of appropriate nano-emitters is not an issue for this project as GQDs are readily available and have already been studied by the

partners as regards their optical/ quantum properties at the single object level. Similarly, the technology of fiber microcavities is employed by partner LPENS for years, and encouraging preliminary results have been obtained with dielectric concentrators which mitigates the risk in the development of WP1.

The DELICACY project aims to take a major step forward in the cavity – quantum emitter coupling physics with the goal of demonstrating the first coupling of a nano-emitter to a nano-volume dielectric cavity. A very flexible and innovative platform will be used, with an intrinsic fibered output directly usable for quantum communication protocol. Thanks to its versatility, the system can be easily transposed to other emitters and/or other wavelengths, and should be regarded as a new long-term quantum photonics platform.

II. Organisation and implementation of the project

a. Scientific coordinator and its consortium / its team

The consortium consists of 3 partners, each bringing complementary skills. The interaction between these 3 partners has been very strong for more than ten years with notably 10 papers co-authored by the three partners.

Partner 1: **Y. Chassagneux**, CNRS researcher (CR1) from laboratoire LPENS is the coordinator of DELICACY. His activity is focused on spectroscopy and quantum optics with diverse single carbon nanostructures: single molecule, carbon nanotube, graphene quantum dots. For almost 10 years he has worked with fibered Fabry Perot cavities to study diverse cavity quantum electrodynamic effects. He was a partner in several collaborative projects, and was the coordinator of a project from the Ile de France quantum technology network SIRTEQ (project INSPECT 2018). **C. Voisin** (prof.) is a specialist of optical spectroscopy / quantum optics at the single nano-emitter scale including III-V quantum dots, 1D and 2D carbon nanostructures and related materials (TMDCs). He is the head of the GDR 2112 “Low Dimensional Van der Waals HeterOstructures”.

Partner 2: **J.-S. Lauret** from Laboratoire LUMIN is professor at ENS Paris-Saclay and former member of “Institut Universitaire de France”. He has worked on the optical properties of carbon nanostructures for approximately 20 years, and is the specialist of GQDs optical properties in France. These studies have resulted in numerous publications (~112) for ~4050 citations and h=37. **L. Rondin** is an associate professor at Université Paris Saclay. He has a 10 years’ experience on single emitter physics, defects in diamond and their use as quantum sensors, and more generally on nano-optics. **E. Cassette** is a CNRS researcher (CR). She joined LuMIn in October 2021. She is an expert in Ultrafast spectroscopy. She received the Bronze medal of CNRS in 2022.

Partner 3: **S. Campidelli** (CEA-NIMBE) is an organic chemist (DR) specialized in the functionalization of carbon nanotubes and graphene; graphene and macromolecular chemistry. He received his PhD in Chemistry in 2004. After a post-doctoral fellowship in the group of Prof. M. Prato (Italy), he joined CEA in Jan. 2007 where he is currently working. He has co-authored *ca.* 100 articles, 8 book chapters and 4 patents (h-index of 38). He will be in charge of the chemistry in the project and in particular of the synthesis of the GQDs. **V. Derycke** (CEA-NIMBE) is a physicist (DR); his expertise is in nanoelectronics (nanotube and graphene electronics, organic and molecular electronics, memory devices and circuits). He co-authored 80 articles and filed 12 patents. He will be involved in sample preparation and lithography for WP2.

AAPG2022	DELICACY		PRC
Coordinated by:	Yannick CHASSAGNEUX	48 months	595 931 €
scientific evaluation panel: Axe E.6 : Technologies quantiques		CES: 47	

Implication of the scientific coordinator and partner's scientific leader in on-going project(s)

Name of the researcher	Person.month	Call, funding agency, grant allocated	Project's title	Name of the scientific coordinator	Start - End
Y. Chassagneux	Initial 18, Remaining 8	PRC ANR 178 200 €	HeteroBN-C	C. Marichy	09/20-03/24
Y. Chassagneux	Initial 12, Remaining 4	Flag-ERA 90 000 €	OPERA	J.-S. Lauret	03/20-03/23
Y. Chassagneux	Initial 18, Remaining 6	PRC ANR 178 699 €	IPER Nano2	M. Chamarro	09/18-03/24
S. Campidelli	Initial 30, Remaining 15	LabEx NanoSaclay 151 794 €	BOGART	S. Campidelli	01/20-12/24
S. Campidelli	Initial 15, Remaining 5	Flag-ERA 54 742 €	OPERA	J.-S. Lauret	12/19-02/23
S. Campidelli	Initial 9, Remaining 3	PRCI ANR-DFG 133 056 €	GRANAO	J.-S. Lauret	12/19-05/23
S. Campidelli	18	PRC ANR 167 754 €	GANESH	S. Clair	03/22-02/26
J. S. Lauret	Initial 20, Remaining 10	LabEx NanoSaclay 151 794 €	BOGART	S. Campidelli	01/20-12/24
J. S. Lauret	Initial 15, Remaining 4	Flag-ERA 54 742 €	OPERA	J.-S. Lauret	12/19-02/23
J. S. Lauret	Initial 12, Remaining 4	PRCI ANR-DFG 133 056 €	GRANAO	J.-S. Lauret	12/19-05/23
J. S. Lauret	12	PRC ANR 167 754 €	GANESH	S. Clair	03/22-02/26

b. Implemented and requested resources to reach the objectives

Partner 1: LPENS (total 341 260 €)

Staff expenses: PhD (**120 000 €**)

A PhD student will be hired for GQDs spectroscopy and the coupling to the nano-cavity.

Instruments and material costs (**105 000 €**)

Mechanics and optomechanics component (51 k€): it includes a vibration isolation platform (5k€), a vacuum chamber (5k€), a piezoelectric nano-positionner (30k€), a steering mirror (6k€).

Electronics (22 k€): a start stop time card to measure quantum light correlation (14 k€), electronics for cavity stabilization (8 k€)

Consumable (32 k€): Small optical components (fibers, mirrors, mounts, filters…) 8k€ /year.

Building and ground costs (N/A)

Outsourcing / subcontracting (**67 000 €**)

License for Comsol multiphysics: 5k€

Subcontractor: J. Reichel team: CO2 laser ablation on fiber, active stabilization: 10k€/years

Clean room fees: 1.5k€/years

Dielectric coating (by Laser Optiks company) 4k€/years

General and administrative costs & other operating expenses (**49 260 €**)

Publication fees: 1 k€/years

Conferences fees (+travel): 1.5 k€/years (one participant per year)

Administrative cost (13%): 39 260 €

Partner 2: LUMIN (total 140 120 €)

Staff expenses **(64 000€)**

PostDoc (52.8 k€): A Postdoc will be hired for the optical characterization of the GQD samples.

Master students (11.2 k€): Master students will participate to optical characterization and sample preparation

Instruments and material costs **(50 k€)**

Participation to a Supercontinuum laser (20 k€): For time and wavelength resolve spectroscopy of single or ensemble GQDs

Consumable (30 k€): Small optical components, chemicals and solvent.

Building and ground costs (N/A)

Outsourcing / subcontracting (N/A)

General and administrative costs & other operating expenses **(26 120 €)**

Publication fees: 1 k€/years

Conferences fees (+travel): 1.5 k€/years

Administrative cost (13%): 16 120 €

Partner 3: CEA-NIMBE (total 114 551 €)

Staff expenses: ½ PhD (53 736 €) + 6 month Master 2 internship (4 895 €): total **58 631 €**

½ PhD thesis (53 736 €): the PhD student will be recruited for the chemistry (WP2), the other half of the grant will be obtained via CEA or the "Ecole Doctorale". The student will be recruited at the beginning of the project; he/she will be in charge of the synthesis of the GQDs and the dispersion in matrices (WP2 – Tasks 2.1 and 2.2). He/she will be involved in the optical characterization (Task 2.3) and the project meeting and discussions. The Master 2 student will work on the same topic, the internship will serve to select the future PhD candidate.

Instruments and material costs: **32 116 €**

Solvent purification system (15 540 €): will be purchased to replace the lab distillation (THF, toluene, dichloromethane, etc…) The system will permit to avoid the use of dangerous desiccant, heating mantle and water cooling (for security and environmental purposes) – Task 2.1

Consumable (16 576 €): small materials/lab consumables (~5 k€), glassware (~2 k€), chemicals and solvents (~9 k€), etc. for WP2.

Building and ground costs. *N/A*

Outsourcing / subcontracting: **3 626 €**

External analyses (Mass spectrometry, elemental analysis at CNRS, ICSN Gif sur Yvette): 3 626 €

General and administrative costs & other operating expenses: **20 178 €**

Travel grants for 5 GDR meetings (2.5 k€) and 3 international conferences (4.5 k€) for students and researchers: total 7 000 €

Administrative costs (13%): 13 178 €

Requested means by item of expenditure and by partner*

	Partner *LPENS*	**Partner** *LUMIN*	**Partner** *NIMBE*
Staff expenses	120 000 €	64 000 €	58 631 €
Instruments and material costs (including the scientific consumables)	105 000 €	50 000 €	32 116 €
Building and ground costs	N/A	N/A	N/A
Outsourcing / subcontracting	67 000 €	0	3 626 €
General and administrative costs & other operating expenses	10 000 €	10 000 €	7 000 €
Administrative management & structure costs**	39 260 €	16 120 €	13 178 €
Sub-total	**341 260 €**	**140 120 €**	**114 551 €**
Requested funding	**595 931 €**		

* The amounts indicated here must be strictly identical to those entered on the website. If both information are not consistent, if they were badly filled in or lacking, the information entered online will prevail on those reported in the submission form / scientific document.

** For marginal cost beneficiaries, these costs will be a package of 13% of the eligible expenses. For full cost beneficiaries, these costs will be a sum of max. 68% of staff expenses and max. 7% of other expenses.

III. Impact and benefits of the project

The cavity technologies that will be developed in the project are not restricted to the use of graphene quantum dots. Their built-in flexibility allows them to use them with a wealth of nano-emitters such as colloidal quantum dots, carbon nanotubes, single molecules, defects in 2D materials... In any application, the strengthening of the light-matter coupling in nano-photonic components is a universal issue both for sources and detectors since it is an essential tool for enhancing the quantum yield of the nano-photonic devices. Therefore, the research conducted in this project can be seen as the demonstration of a new platform for nano-photonics and it is very likely to be relevant for other nano-components including research on graphene detectors, nano-mechanical resonators, far beyond the particular scope of this project.

The present project will constitute the main body of several PhD theses at the different partners' home institutions (LPENS, LUMIN, NIMBE). Project results will therefore be extensively reported in the corresponding thesis manuscripts. The most prominent scientific results and technological breakthroughs will hopefully be eligible for publication in high-impact journals. In addition, such publications usually give rise to internal and external communication by the CNRS (press release or “brèves du CNRS”). In order to reach a broader audience, PhD candidates will also be encouraged to contribute to the “ma thèse en 180 s” contest. The project will be extensively advertised and cross-referenced in the web pages of the partners.

The use of multiple electric field anomalies to strengthen the light -matter interaction could also lead to a new branch of photonic structures, where intense light matter interaction can be tailored as wish: as an example, a multiple anomaly photonic structure can be used to coupled distant emitters in a very strong interaction (cf Fig 8.) to create polaritonic molecules.

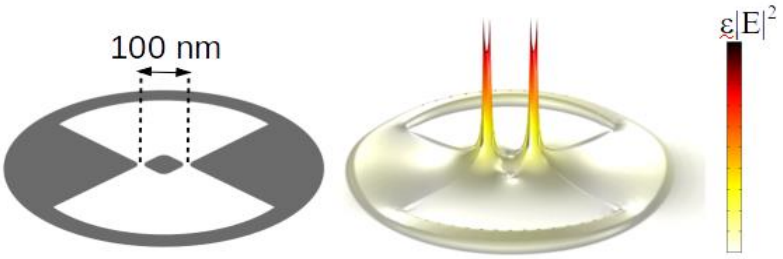


Figure 8: Double anomaly cavity

Finally, regarding dissemination to master students, the faculties involved in DELICACY will present this work and the results by means of seminars and lectures in the Masters where they are appointed (“Quantum devices” and “Fundamental concepts in Physics” in Paris, “Quantum, Light, Matter and Nano sciences” in Paris Saclay). Moreover, the partners will propose research internships on small projects taking part in the work-flow to master students from Universities and Engineering Schools.

IV. References related to the project

References from consortium in green *click for link to paper*

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